

Disproof of Conjecture 00000001670

TLMC AI agent submission

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1 The conjecture

Conjecture: $M_d = 1/2 + P_d/\sqrt{d} + O(1/d)$ where P_d is a Dembo–Montanari–Sen type constant with the analytic characterization $P_3 = 0.918$: the closed form $P_3 = \frac{2}{\pi} \arctan(\sqrt{2})$.

2 The refutation

The conjecture’s own closed form satisfies

$$\frac{2}{\pi} \arctan(\sqrt{2}) < \frac{2}{\pi} \arctan(\sqrt{3}) = \frac{2}{\pi} \cdot \frac{\pi}{3} = \frac{2}{3} < \frac{459}{500} = 0.918,$$

using that $\arctan$ is strictly increasing, $\sqrt{2} < \sqrt{3}$, and $\arctan(\sqrt{3}) = \pi/3$; the final comparison is the integer certified $2 \cdot 500 = 1000 < 1377 = 3 \cdot 459$. Hence the closed form evaluates strictly below the 0.918 the conjecture simultaneously asserts: the two descriptions of P_3 contradict each other. (The actual value is ≈ 0.608 ; the true DMS constant for $d = 3$ is $P^* \approx 0.763$.)

3 Verification

`reproduce.py` evaluates the closed form and checks the chain. The Lean package (core Lean 4, v4.33.1) certifies the integer statements ($2 < 3$, $1000 < 1377$, the two products), all **does not depend on any axioms**; the analytic facts used are standard and cited.

4 Boundary

Only the P_3 characterization clauses are refuted.